Dry runs, trace tables, data requirements and interfaces

D13 FACT FILE

Specification: students should be able to

- Use a dry run to test a solution
- Identify data requirements for a solution and develop an appropriate user interface

CCEA fact file: Digital Design Principles

Key words: dry run, trace table, desk check, data requirements, variable, data dictionary, user interface, prompt

```
a ← 1
WHILE a < 20
    a ← a * 2
END WHILE
OUTPUT a
```

Check the condition every time, including the last one: 32 < 20 is False, so the loop stops and 32 is output – not 16.

a	a < 20 ?	Output
1	True	
2	True	
4	True	
8	True	
16	True	
32	**False**	**32**

A dry run of a WHILE loop recorded in a trace table

Dry runs and trace tables

A dry run means stepping through an algorithm by hand, keeping track of every variable in a trace table and writing down any output. It checks the logic is correct and that the expected output is produced, without needing a computer. Pay particular attention to loop conditions: check the condition every time, including the final time when it becomes false.

Data requirements and interfaces

A designer must identify the data the solution needs – inputs, processed values and outputs – along with sensible names, data types and validation rules. The user interface should give clear prompts and helpful messages, be laid out logically, and suit the intended user.

T Key terms from the mark schemes

Dry run
Working through code by hand, line by line, recording variable values.
Credited: U4 2019 Q10, 2022 Q8, 2025 Q5

Trace table
Table recording the values of variables at each step of a dry run.
Credited: U4 2024 Q7, 2025 Q5

Desk checking
Reviewing code by hand to find errors before running it.
Credited: U4 2025 Q5

C C# example

Predicting loop output

```
for (int i = 5; i < 8; i++)
{
    Console.WriteLine(i * 3);
}
```

Exam focus

Trace and predict questions appear on every paper: 2019 Q10(a) table, 2022 Q8(b) dry runs, 2023 Q2, 2024 Q1–Q2, 2025 Q2 and Q5 (dry run explanation, trace table, final value).

Unit 4 2025 Q5	[7]	Desk checking, trace table, final value
Unit 4 2025 Q2	[12]	Loop output, nested loops, loop count, infinite loop, fixing a condition
Unit 4 2024 Q2	[3]	Tracing FOR and WHILE loops
Unit 4 2024 Q1	[4]	Dry-run a nested IF (shape calculator)
Unit 4 2023 Q2	[4]	Tracing nested selection

? Check yourself

1. Explain how a dry run helps when testing a program. [2]
2. What is a trace table? [1]
3. Predict the output of: for (int i = 1; i < 4; i++) { Console.WriteLine(i * i); } [2]

Answers: 1. Working through the algorithm/program by hand, line by line, without a computer (1) / to find logic errors and check the output is as expected (checking every path) (1) 2. A table that records the value of each variable (and the condition and any output) after each step of an algorithm (1) 3. The loop runs for i = 1, 2, 3 (it stops when i = 4 because i < 4 is false) (1) / output 1, 4, 9, each on a new line, and nothing else (1). No credit for the second mark if 16 is included.